

NAVLE High-Yield Infectious Diseases

Top 5 — dogs & cats • Top 3 — horses, cattle, swine, sheep/goats, exotics, fish

How to use: the last column carries the exam bait — buzzwords, pathognomonic findings, and **zoonotic / reportable** flags. NAVLE loves three things: (1) the classic buzzword-to-agent link, (2) drug of choice, and (3) “who do I have to call?” Any vesicular oral/foot lesion in livestock, sudden high-mortality event, or suspected foreign animal disease → contact your State Animal Health Official / USDA APHIS AVIC.

DOGS — Top 5

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Canine parvovirus (CPV-2) Non-enveloped DNA virus; fecal-oral	Unvaccinated pup 6 wk–6 mo. Rottweiler/Dobie/Pit predisposed. Foul, hemorrhagic diarrhea, vomiting, lethargy, fever → hypothermia. Targets crypt cells + bone marrow.	Fecal antigen ELISA (SNAP). CBC: panleukopenia (neutropenia + lymphopenia). Vaccine can cause weak false-positive 5–12 d post-vax.	Supportive: IV crystalloids ± colloids, dextrose, K⁺ ; maropitant/ondansetron; broad-spectrum abx (ampicillin ± enrofloxacin) for sepsis from neutropenia; early enteral nutrition. Monoclonal antibody (CPMA) available. Bleach 1:32.	Buzzword: parvo + intussusception. Myocarditis form in neonates. Vaccinate q3–4 wk to ≥16 wk. Virus survives months in environment; quaternary ammonium does NOT kill it.
Canine distemper virus Enveloped morbillivirus (paramyxo); aerosol	Biphasic fever, oculonasal discharge, pneumonia, GI signs, then CNS: myoclonus (“chewing gum fits”), seizures, ataxia. Hyperkeratosis of nose/footpads ; enamel hypoplasia if infected pre-eruption.	PCR (conjunctival swab, whole blood, CSF, urine). Inclusion bodies in RBCs/lymphocytes. CSF antibody titer. Lymphopenia.	No specific antiviral — supportive care, anticonvulsants. Prevent with MLV core vaccine. Enveloped → labile, easily killed by routine disinfectants.	“Hard pad disease.” Old-dog encephalitis. CNS signs may appear weeks after recovery. Also infects ferrets (high mortality) and wildlife.
Leptospirosis <i>Leptospira interrogans</i> serovars; spirochete	Large-breed outdoor dog, rain/flood exposure, rodent or wildlife urine. Acute kidney injury ± hepatic injury, fever, myalgia (reluctance to move), PU/PD then oliguria. Pulmonary hemorrhage syndrome.	MAT paired titers (4× rise) — vaccine cross-reacts (titers usually ≤1:800). PCR blood (early) + urine (later). Chem: azotemia, ↑ALP, hypoK, hypoNa. Glucosuria without hyperglycemia.	IV ampicillin/penicillin for acute phase, then doxycycline 2 wk to clear the renal carrier state . Fluids/dialysis for AKI. Bacterin vaccine (4-serovar).	ZOONOTIC — gloves, avoid urine contact, warn owner. Reportable in some states. Classic combo: AKI + ↑liver enzymes in a young dog.
CIRDC / “kennel cough” <i>Bordetella bronchiseptica</i>, parainfluenza, CAV-2, <i>Mycoplasma</i>, canine influenza	Recent shelter/boarding/daycare. Loud honking cough ± terminal retch, easily induced by tracheal palpation. Bright, afebrile, eating normally in uncomplicated cases.	Usually clinical + history. PCR respiratory panel for outbreaks. Thoracic rads if febrile/anorexic to rule out pneumonia.	Self-limiting in 1–2 wk; cough suppressant (hydrocodone/butorphanol), rest, harness not collar. Doxycycline if febrile or purulent. Intranasal/oral <i>Bordetella</i> + CPiV vaccines give fastest mucosal immunity.	Buzzword: bright dog with dramatic cough. Intranasal vaccine may cause mild transient cough. Zoonotic risk in immunocompromised people.
Lyme disease <i>Borrelia burgdorferi</i>; <i>Ixodes scapularis/pacificus</i>	Northeast/upper Midwest/Pacific coast. Shifting/intermittent lameness , fever, regional lymphadenopathy. Lyme nephritis (protein-losing nephropathy) in Labs/Goldens — grave.	C6 peptide (SNAP 4Dx) — distinguishes infection from vaccination ; confirm with quantitative C6. Joint tap: neutrophilic. Screen UPC in seropositive dogs.	Doxycycline 10 mg/kg q12h × 4 wk — dramatic response in 24–48 h. Treat only seropositive dogs that are clinical or proteinuric. Tick prevention (isoxazolines) + Lyme vaccine in endemic areas.	Tick must feed = 48 h. No erythema migrans rash in dogs. Do NOT expect seroreversion after treatment. Vector-borne differentials: <i>Ehrlichia</i> , <i>Anaplasma</i> , RMSF, <i>Babesia</i> .

CATS — Top 5

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Feline panleukopenia (FPV) Feline parvovirus	Unvaccinated kitten: fever, vomiting, diarrhea, dehydration, thickened intestinal loops. In utero / neonatal infection → cerebellar hypoplasia (ataxia, intention tremor, wide stance; non-progressive).	Fecal canine parvo ELISA cross-reacts. CBC: profound panleukopenia. Recent MLV vaccine → false positive.	Aggressive supportive care, antiemetics, abx for sepsis. Bleach. Core MLV vaccine — use killed in pregnant queens/kittens <4 wk (MLV causes cerebellar hypoplasia).	Buzzword: kitten with ataxia but bright and normal mentation = cerebellar hypoplasia. Very high mortality untreated.
Feline leukemia virus (FeLV) Gammaretrovirus; saliva, “friendly cat” spread	Young cat, outdoor/multi-cat. Lymphoma (mediastinal in young cats), non-regenerative anemia, myelosuppression, secondary infections, reproductive failure.	ELISA / SNAP detects p27 antigen (blood, not tears/saliva). Positive → retest in 30 d and confirm with IFA (bone-marrow stage) or PCR for provirus. No maternal-antibody problem (antigen test).	No cure. Supportive care, keep indoors, treat secondary disease. Vaccinate at-risk cats — test before vaccinating. Non-core in indoor-only adults.	Progressive vs regressive infection. FeLV■ cats can outlive expectations. Buzzword: mediastinal lymphoma in a young cat = FeLV .
Feline immunodeficiency virus (FIV) Lentivirus; bite wounds	Adult intact outdoor male tomcat. Gingivostomatitis, chronic rhinitis, weight loss, opportunistic infection, lymphoma. Long asymptomatic phase — many live normal lifespans.	ELISA detects antibody → kittens <6 mo may have maternal antibody (retest after 6 mo); previously vaccinated cats test positive. Confirm with Western blot or PCR.	No cure. Neuter, keep indoors, semi-annual exams, dental care, no MLV vaccines in symptomatic cats. FIV vaccine no longer marketed in the US.	Not zoonotic. Transmission is inefficient by casual contact — FIV■ cats can often live with non-aggressive housemates.
FIP Virulent biotype of feline coronavirus (FCoV)	Cat <2 yr from multi-cat/shelter, purebred. Wet form : straw-colored, high-protein abdominal/pleural effusion, fever unresponsive to abx. Dry form : granulomas, uveitis, retinal vasculitis, neuro signs .	No single test. Support: hyperglobulinemia, A:G ratio <0.4 , non-regenerative anemia, ↑bilirubin without hemolysis. Effusion: Rivalta positive , protein >3.5 g/dL, low cellularity. Definitive = immunostaining of FCoV antigen in macrophages / histopath. FCoV serology does NOT diagnose FIP.	GS-441524 antiviral — legally compounded in the US under FDA enforcement discretion (2024); oral route preferred; ≈84 d courses (shorter protocols emerging). Formerly uniformly fatal.	Buzzword: young cat + fever + effusion + low A:G. Pyogranulomatous vasculitis is the lesion. Not contagious as FIP — the enteric FCoV is what spreads.
Feline URI complex FHV-1 (alphaherpes) + feline calicivirus	FHV-1 : sneezing, serous→mucopurulent discharge, conjunctivitis, dendritic corneal ulcers , symblepharon; latency in trigeminal ganglion → stress recrudescence. FCV : oral/lingual ulcers , limping syndrome; virulent systemic FCV = edema, ulcerative dermatitis, high mortality.	PCR (conjunctival/oropharyngeal) — interpret cautiously (carriers, post-vaccination). Fluorescein: branching dendritic ulcer = FHV-1 pathognomonic .	Supportive: nebulization, appetite stimulants (mirtazapine), clean nares. FHV-1: topical/oral famciclovir , idoxuridine/cidofovir; L-lysine of doubtful benefit. Doxycycline if <i>Chlamydia</i> / <i>Mycoplasma</i> suspected. Core FVRCP vaccine reduces severity, not infection.	Never use topical corticosteroids on a fluorescein-positive eye. Dendritic ulcer = herpes; oral ulcers = calici. FCV is non-enveloped → needs bleach/accelerated H ₂ O ₂ .

HORSES — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Strangles <i>Streptococcus equi</i> subsp. <i>equi</i>	Young horse, new arrival/show. High fever, mucopurulent nasal discharge, submandibular/retropharyngeal lymphadenopathy that abscesses and ruptures , dysphagia, extended head/neck. Sequelae: guttural pouch empyema/ chondroids , purpura hemorrhagica , “bastard strangles.”	Culture or PCR of nasopharyngeal lavage / guttural pouch lavage (best for carrier detection). Endoscopy of guttural pouch. Fever screening is the earliest outbreak sign.	Hot-pack and lance mature abscesses; penicillin (procaine pen G IM) — controversial early (may blunt immunity, may abort abscess). Isolate 3 wk; 3 negative GP lavages to release. Quarantine new arrivals 3 wk.	Reportable in many states . Purpura hemorrhagica = immune-complex vasculitis, treat with corticosteroids. Guttural pouch is the carrier site — must clear chondroids.

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Equine herpesvirus-1 (EHV-1) Alphaherpesvirus (EHV-4 mostly respiratory only)	Fever, biphasic; respiratory disease in weanlings; late-gestation abortion storms (no premonitory signs); EHM — myeloencephalopathy: ataxia worse in hindlimbs, urinary incontinence/bladder atony , tail/perineal hypalgesia, dog-sitting.	PCR of nasal swab + whole blood (buffy coat) . Aborted fetus/placenta for PCR + histopath (hepatic necrosis, inclusion bodies). CSF: xanthochromia, high protein.	Supportive; valacyclovir , NSAIDs, corticosteroids for EHM (controversial), sling/bladder catheterization. Strict isolation, twice-daily temps, disinfect (enveloped → labile). Vaccinate broodmares at 5, 7, 9 mo gestation.	EHM is reportable in most states. Latency in trigeminal ganglion + lymphoid tissue → shedding with stress. Neuropathogenic D₇₅₂ strain.
Equine infectious anemia (EIA) Lentivirus; horsefly/deerfly mechanical, contaminated needles	"Swamp fever." Acute: fever, thrombocytopenia, petechiae, epistaxis. Chronic: recurrent fever, weight loss, ventral edema , anemia, icterus. Most become inapparent lifelong carriers.	Coggins test = AGID (gold standard); cELISA is more sensitive/faster — confirm positives with AGID. Foals: maternal antibody until ~6–12 mo.	No treatment, no vaccine. Positive horses: lifelong quarantine ≥200 yards from other equids, or euthanasia. Single-use needles.	FEDERALLY REPORTABLE. Negative Coggins required for interstate travel, sales, shows. Buzzword: ventral edema + recurrent fever + thrombocytopenia.

CATTLE — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Bovine respiratory disease / shipping fever <i>Mannheimia haemolytica</i> (± <i>Pasteurella multocida</i>, <i>Histophilus somni</i>, <i>Mycoplasma bovis</i>) after BRSV/IBR/BVDV stress	Recently weaned/shipped feedlot calf. Fever >104°F, depression, drooped ears, extended head, crackles, nasal discharge; cranioventral consolidation .	Clinical + rectal temp + lung auscultation. Transtracheal wash/BAL for culture + sensitivity. Necropsy: fibrinous bronchopneumonia, cranioventral.	Florfenicol, tulathromycin, ceftiofur, enrofloxacin + NSAID (flunixin). Metaphylaxis on arrival for high-risk groups. Preconditioning: vaccinate + wean 45 d before shipping, low-stress handling.	Buzzword: cranioventral lung consolidation = bacterial; caudodorsal = viral/interstitial or atypical. Watch withdrawal times and extralabel drug rules (AMDUCA); fluoroquinolone use in cattle is prohibited extralabel.
Bovine viral diarrhea virus (BVDV) Pestivirus (cytopathic + noncytopathic)	Reproductive failure, immunosuppression, oral erosions, diarrhea. PI calf : infected 40–125 d gestation — poor-doing, persistent shedder. Mucosal disease : PI calf superinfected with cytopathic strain → profuse diarrhea, oral/coronary band erosions, 100% fatal. Also cerebellar hypoplasia, ocular defects.	Ear-notch antigen-capture ELISA or IHC for PI screening (whole herd). PCR on pooled ear notches/milk. Serology of unvaccinated calves for exposure.	Identify and cull PI animals — the cornerstone. Biosecurity, test purchases. Vaccinate (MLV pre-breeding; killed in pregnant naive cattle).	Buzzword: young adult with erosive oral lesions + diarrhea, unresponsive → mucosal disease. Differential for FMD/vesicular stomatitis — both REPORTABLE ; call state vet for any vesicular oral lesion.
Johne's disease (paratuberculosis) <i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> (MAP)	Adult cow >2 yr (infected as a calf via feces/colostrum). Chronic, progressive, watery "pipe-stream" diarrhea with a NORMAL appetite and NO fever ; weight loss, submandibular edema ("bottle jaw") from protein loss.	Fecal PCR (or culture, slow) = most sensitive; serum ELISA for herd screening (poor in early disease). Necropsy: corrugated/thickened ileum , enlarged mesenteric nodes; acid-fast clumps in smears.	No effective treatment — cull positives and their offspring. Calve in clean areas, remove calves from dams immediately, pasteurized colostrum/milk replacer, don't spread manure on calf pastures.	Iceberg disease: 1 clinical = many subclinical. Debated (unproven) link to human Crohn's. Differentials for chronic adult diarrhea: salmonellosis, ostertagiasis, copper deficiency, winter dysentery .

SWINE — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
PRRS Arterivirus ("blue-ear disease")	Sows : late-term abortion, stillborns, mummies, weak pigs, anorexia, transient ear cyanosis. Nursery/grow-finish : interstitial pneumonia, poor growth, secondary bacterial pneumonia (PRDC with <i>M. hyopneumoniae</i> , <i>Strep suis</i>).	PCR (serum, oral fluids — rope sampling, processing fluids), ELISA serology, sequencing (ORF5) for strain matching. Lung histopath: interstitial pneumonia.	No specific treatment. Abx for secondary bacteria. Control: MLV or killed vaccine, herd closure + rollover, McREBEL, all-in/all-out, air filtration, strict biosecurity/transport hygiene.	Most economically costly US swine disease. Not zoonotic. Vaccine protection is strain-dependent — hence sequencing.

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Swine erysipelas <i>Erysipelothrix rhusiopathiae</i>	Acute: high fever, raised, rhomboid “diamond-skin” lesions , sudden death. Chronic: proliferative arthritis and vegetative valvular endocarditis (mitral) → lameness, poor growth, sudden death.	Clinical diamond lesions are near-pathognomonic. Culture from blood/joint/valve. Response to penicillin supports diagnosis.	Penicillin — dramatically responsive (rapid, near-diagnostic response). Bacterin vaccine for breeding herd and grow-finish. Tonsillar carriers shed in feces.	ZOONOTIC — “erysipeloid”/fish-handler’s disease, cellulitis from skin puncture in swine handlers, butchers, fishermen. Also affects turkeys and sheep.
Porcine epidemic diarrhea virus (PEDV) / TGE Alphacoronavirus	Watery diarrhea + vomiting sweeping through all ages ; near-100% mortality in piglets <1 wk (villous atrophy → malabsorption); adults recover in ~1 wk. Dehydration, huddling.	PCR on feces/oral fluids; IHC/histopath of small intestine showing severe villous atrophy . Differentiate PEDV vs TGE vs porcine deltacoronavirus vs rotavirus by PCR.	Supportive fluids/electrolytes, warmth. Feedback/planned exposure of pregnant sows for lactogenic immunity; strict biosecurity, dry time and heat on transport trailers, feed-ingredient risk control.	Entered the US in 2013. Always keep foreign animal diseases on the list for sick pigs: African swine fever, classical swine fever, FMD, pseudorabies — REPORTABLE, call state/federal vet immediately. ASF: high fever, cyanotic skin, splenomegaly, high mortality.

SHEEP & GOATS — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Caseous lymphadenitis (CL) <i>Corynebacterium pseudotuberculosis</i>	Firm abscesses at superficial lymph nodes (parotid, submandibular, prescapular, prefemoral) with thick, pasty, greenish-white, odorless pus (“toothpaste”). Internal form: chronic wasting, respiratory signs (“thin ewe syndrome”).	Aspirate + culture. Synergistic hemolysis-inhibition test/ELISA for internal form (vaccination confounds). Ultrasound for internal abscesses.	Cull or isolate. If treating: lance/flush when mature away from other animals, dispose of pus and bedding, or surgical excision en bloc. Antibiotics penetrate the capsule poorly. Vaccines available for sheep (can cause reactions in goats). Shear/handle young clean animals first; treat wounds.	ZOONOTIC (rare). Organism persists in soil/bedding for months. Pus draining into the environment perpetuates the herd problem — that’s why culling wins.
CAE / OPP Small-ruminant lentiviruses (caprine arthritis-encephalitis / ovine progressive pneumonia “maedi-visna”)	Goats: chronic proliferative arthritis (“big knee” — carpi), hard/agalactic udder , weight loss; encephalomyelitis with ascending paresis in 2–6 mo kids. Sheep: progressive dyspnea/wasting with no fever, no cough response to abx ; visna = neuro form.	AGID or ELISA serology (whole-herd, annual). PCR. Rads of carpi; lung histopath: lymphoid interstitial pneumonia. Seroconversion can take months.	No treatment, no vaccine. Control: serologic test-and-cull, remove kids/lambs at birth and feed heat-treated colostrum (56°C × 60 min) or pasteurized milk , separate positive/negative groups, closed herd.	Transmission mainly via colostrum/milk, also horizontal with close contact. Not zoonotic. Buzzword: adult goat, big knees + hard udder = CAE.
Contagious ecthyma (“orf” / soremouth) Parapoxvirus	Proliferative, crusty, scabby lesions at commissures of the lips, muzzle, nostrils ; also teats of the dam (→ mastitis, rejection of lamb) and coronary bands. Lambs/kids reluctant to nurse → poor growth.	Clinical appearance is characteristic. Electron microscopy, PCR, or histopath of scab if needed. Differentiate from bluetongue, FMD, and sheep/goat pox (foreign, reportable) .	Self-limiting in 1–4 wk. Soften scabs, treat secondary bacterial infection, ensure nursing/nutrition; tube-feed if needed. Live scab vaccine (scarification) only in already-infected flocks — it introduces the virus.	ZOONOTIC — wear gloves. Painful nodule on the human hand/finger. Also know the small-ruminant abortion trio: Chlamydia abortus (EAE), Campylobacter, Toxoplasma — all with zoonotic risk; C. abortus and Q fever are dangerous to pregnant women. And scrapie — REPORTABLE (pruritus, wool loss, ataxia; official ID/flock certification).

EXOTIC / COMPANION ZOO SPECIES — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Psittacosis / avian chlamydiosis <i>Chlamydia psittaci</i> (birds)	Recently acquired or stressed psittacine (cockatiels, budgies, Amazons). Lethargy, ruffled feathers, bright lime-green urates , biliverdinuria, conjunctivitis, sinusitis, dyspnea, diarrhea. Hepatosplenomegaly.	PCR of combined choanal/cloacal swab (best antemortem); serology + antigen tests as adjuncts; impression smears (Gimenez/Macchiavello). CBC: leukocytosis; ↑AST/bile acids.	Doxycycline 45 days (oral, medicated feed, or IM long-acting). Treat all exposed birds, disinfect (labile — quats, bleach), quarantine new birds 30–45 d. Wear N95 + gloves when cleaning.	ZOONOTIC and REPORTABLE in most states. Human disease = atypical pneumonia, fever, headache; can be severe. Also flag: HPAI and virulent Newcastle disease — FEDERALLY REPORTABLE foreign animal diseases.
“Snuffles” / rabbit pasteurellosis <i>Pasteurella multocida</i> (± <i>Bordetella</i> , <i>Staphylococcus</i>)	Pet rabbit: serous → mucopurulent nasal and ocular discharge, matted forepaws from grooming, sneezing, dacryocystitis, head tilt (otitis media/interna), subcutaneous abscesses, pneumonia. Often triggered by poor ventilation/ammonia, crowding, stress.	Deep nasal culture + sensitivity (nares are often contaminated — culture deep or from an abscess). Skull/thoracic rads or CT for otitis, dental disease, and lung abscesses. Rule out dental disease and dacryocystitis — very common mimics.	Enrofloxacin, trimethoprim-sulfa, or azithromycin ; long courses; abscesses need excision (rabbit pus is caseous and does not drain). Fix husbandry: ventilation, low-dust bedding, unlimited hay.	NEVER give oral penicillins, ampicillin, amoxicillin, clindamycin, or lincomycin to rabbits — fatal dysbiosis/enterotoxemia (<i>Clostridium spiriforme</i>). Also know: rabbit hemorrhagic disease virus 2 (RHDV2) — REPORTABLE , sudden death with hepatic necrosis and epistaxis; <i>Encephalitozoon cuniculi</i> (head tilt, cataracts, renal).
Reptile salmonellosis <i>Salmonella</i> spp. (turtles, bearded dragons, snakes)	Usually a normal, intermittent gut commensal in a clinically healthy reptile . Can cause septicemia, hepatitis, osteomyelitis, coelomitis in stressed/immunosuppressed animals kept at incorrect temperature.	Fecal/cloacal culture — but a positive culture in a healthy reptile is expected and does not indicate disease ; intermittent shedding means negatives don't rule out carriage. Blood culture if septic.	Do not attempt to eliminate carriage with antibiotics — ineffective and promotes resistance. Treat clinical illness based on culture/sensitivity + correct the POTZ (preferred optimal temperature zone), humidity, and nutrition.	ZOONOTIC — the key client-education point. No reptiles/amphibians in households with children <5 yr, elderly, pregnant, or immunocompromised people; handwashing; no kitchen sinks/bathubs for cleaning enclosures. US ban on sale of turtles with shell <4 inches. Also flag: canine distemper and influenza in ferrets; <i>Mycobacterium</i> in many exotics.

FISH — Top 3

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
“Ich” / white spot <i>Ichthyophthirius multifiliis</i> (freshwater ciliated protozoan)	Discrete white spots like grains of salt on skin/fins/gills; flashing, rubbing on objects, piping at the surface, lethargy, increased respiration. Often follows a new addition or a temperature drop. Rapid, high mortality in crowded systems.	Wet mount / skin scrape or gill biopsy — look for the large, dark, horseshoe-shaped macronucleus and rotating ciliated trophont. Diagnose on the live fish, not after death.	Formalin, copper sulfate (freshwater), or sodium chloride baths/prolonged immersion; repeat treatments to catch the free-swimming theront (only this stage is drug-susceptible). Raise temperature to speed the life cycle. Quarantine new fish; check ammonia/nitrite/nitrate, DO, temperature.	The encysted trophont under the epithelium is protected — hence repeat dosing. Marine analog = <i>Cryptocaryon irritans</i> . Always evaluate water quality first — it is the underlying cause in most fish cases.
Columnaris <i>Flavobacterium columnare</i> (Gram-negative)	Warm-water fish. “Saddleback” depigmented dorsal lesion, frayed/eroded fins, yellow-brown gill necrosis, mouth “cotton-wool” plaques. Triggered by high temperature, high organic load, crowding, handling injury.	Wet mount of lesion/gill: long, thin Gram-negative rods forming “haystacks” or columns with flexing motility. Culture on selective (Orda's/Hsu-Shotts) medium.	Oxytetracycline or florfenicol in feed (FDA-approved for food fish — mind withdrawal times); potassium permanganate or salt baths for external lesions. Lower temperature and stocking density, improve water quality and filtration.	Often mistaken for a fungus — it is bacterial. Other core bacteria: <i>Aeromonas</i> (ulcers, dropsy), <i>Vibrio</i> (marine), <i>Edwardsiella ictaluri</i> (“hole in the head” in catfish), <i>Streptococcus iniae</i> (— zoonotic via handling injury).

Disease / Agent	Key Signs & Signalment	Diagnosis	Treatment / Control	NAVLE Flags & Buzzwords
Saprolegniasis <i>Saprolegnia</i> spp. (water mold / oomycete)	White-to-gray cotton-wool tufts on skin, fins, gills, or eggs. Almost always secondary to handling trauma, low temperature, poor water quality, or another primary infection. Also destroys eggs in hatcheries.	Wet mount: broad, non-septate, branching hyphae . Note whether it is on damaged tissue — that tells you to hunt for the primary insult.	Salt, potassium permanganate, formalin, or hydrogen peroxide baths; fix the primary cause — water quality, netting/handling technique, temperature, stocking density. Remove dead fish and eggs.	Opportunistic — treating the mold without correcting husbandry guarantees relapse. Also know the REPORTABLE/WOAH-listed viral diseases: viral hemorrhagic septicemia (VHS), spring viremia of carp (SVC), koi herpesvirus (KHV), infectious salmon anemia — report to state and federal authorities.